Oil System – General Data

1. General

CAUTION **DO NOT MIX DIFFERENT BRANDS OF OIL WHEN CHANGING OR ADDING OIL BETWEEN OIL CHANGES, SINCE THE CHEMICAL STRUCTURE OF DIFFERENT BRANDS OF SYNTHETIC OIL MAY DIFFER SUFFICIENTLY TO MAKE THEM INCOMPATIBLE.**

Each engine is equipped with an integral oil tank. The filler cap is accessible through an access panel in the top of the nacelle structure. The total oil capacity for each tank is 1.9 Imp (2.3 U.S.) gallons and the usable oil capacity is 5 Imp (6 U.S.) quarts. The approved synthetic lubricating oils are listed in Pratt and Whitney Canada SB No. 1001.

2. Checking Oil Quantity

A quantity dipstick is contained in the filler cap and indicates the quantity of oil required to top up tank to the full mark. To check oil quantity, open panel to gain access to oil filler cap, remove cap and check oil level in normal manner. Install and secure cap on completion of check.

Note Take care to avoid overfilling the oil tank, which may cause discharge through breather and high oil consumption.

If level indication is low when checking oil quantity after engine has been shutdown for more than 30 minutes, or after an oil change, then start the engine and run at ground-idle for minimum five minutes, feather the propeller, shutdown the engine, and recheck the oil quantity.

3. Draining Oil Systems

To drain the oil system, open the lower engine cowling and remove the drain plugs from the bottom of the accessory gear case, oil cooler, reduction gearbox and oil tank.

4. Oil System Flushing

If engine is to be operated on a different oil than the oil on which it was previously operated, proceed as follows:

A. Remove oil drain plugs from engine oil tank, reduction gear box, accessories gearbox and oil cooler.

CAUTION **LIMIT ENGINE ROTATION TO SHORTEST TIME NECESSARY TO ACCOMPLISH COMPLETE DRAINING.**

B. With drains open, place power lever to IDLE and fuel lever to OFF, motor engine with starter only (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data) to allow scavenge pumps to clear engine.

C. Install drain plugs in engine and oil cooler.

D. Fill engine oil tank with the new oil.

E. Start engine and run at low IDLE for a minimum of two minutes. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

F. Feather propeller.

G. Shutdown Engine. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

H. Repeat Para 4.A and Para 4.B.

I. Remove engine oil filter. Install new element.

J. Reinstall all drain plugs in engine and cooler.

K. Replenish engine oil tank with applicable oil.

L. Start engine. Run at low IDLE for a minimum of two minutes and check for leaks.

M. Feather propeller.

N. Shutdown engine and refill oil tank to proper level.